Table 1: Retention under streaming updates across four datasets, reported with FinalAvg ↑, Forgetting ↓, and BWT ↑. ↑ / ↓ indicate higher / lower is better. Results are shown as mean ± standard deviation over 5 random seeds. **Bold** and <u>underline</u> denote the best and second-best results, respectively.

Dataset / Metric		Continual	Replay		Regularization		Projection		Ours	Improvement
		LoRA	iCaRL	A-GEM	LwF	EWC	OGD	GPM	**SCAU-Rec**	**vs. Best**
MovieLens	FinalAvg ↑	0.2260 ± 0.019	0.2770 ± 0.006	0.2870 ± 0.010	0.3200 ± 0.015	0.3410 ± 0.009	0.3740 ± 0.013	<u>0.3940 ± 0.011</u>	**0.4300 ± 0.005**	+9.1%
	Forgetting ↓	0.2700 ± 0.013	0.2161 ± 0.020	0.2026 ± 0.018	0.1756 ± 0.011	0.1621 ± 0.014	0.1215 ± 0.002	<u>0.0946 ± 0.003</u>	**0.0790 ± 0.011**	+16.5%
	BWT ↑	-0.2161 ± 0.007	-0.1898 ± 0.018	-0.1620 ± 0.017	-0.1405 ± 0.005	-0.1297 ± 0.009	-0.0968 ± 0.012	<u>-0.0756 ± 0.023</u>	**-0.0640 ± 0.009**	+15.3%
Steam	FinalAvg ↑	0.2330 ± 0.014	0.2880 ± 0.005	0.2980 ± 0.018	0.3370 ± 0.007	0.3580 ± 0.011	0.3920 ± 0.003	<u>0.4200 ± 0.016</u>	**0.4490 ± 0.008**	+6.9%
	Forgetting ↓	0.2600 ± 0.004	0.2080 ± 0.017	0.1951 ± 0.003	0.1690 ± 0.014	0.1559 ± 0.006	0.1170 ± 0.019	<u>0.0910 ± 0.008</u>	**0.0752 ± 0.012**	+17.2%
	BWT ↑	-0.2080 ± 0.021	-0.1664 ± 0.006	-0.1559 ± 0.015	-0.1351 ± 0.003	-0.1246 ± 0.018	-0.0936 ± 0.009	<u>-0.0726 ± 0.004</u>	**-0.0593 ± 0.017**	+18.3%
LastFM	FinalAvg ↑	0.2510 ± 0.008	0.3050 ± 0.012	0.3220 ± 0.003	0.3630 ± 0.019	0.3780 ± 0.006	0.4140 ± 0.011	<u>0.4370 ± 0.004</u>	**0.4580 ± 0.015**	+4.8%
	Forgetting ↓	0.2500 ± 0.017	0.2000 ± 0.005	0.1876 ± 0.021	0.1626 ± 0.004	0.1500 ± 0.009	0.1126 ± 0.016	<u>0.0876 ± 0.013</u>	**0.0758 ± 0.003**	+13.5%
	BWT ↑	-0.2042 ± 0.004	-0.1650 ± 0.016	-0.1500 ± 0.008	-0.1300 ± 0.011	-0.1200 ± 0.020	-0.0900 ± 0.003	<u>-0.0700 ± 0.007</u>	**-0.0614 ± 0.018**	+12.3%
Goodreads	FinalAvg ↑	0.2060 ± 0.010	0.2520 ± 0.013	0.2730 ± 0.006	0.2920 ± 0.018	0.3120 ± 0.009	0.3420 ± 0.012	<u>0.3610 ± 0.007</u>	**0.4080 ± 0.013**	+13.0%
	Forgetting ↓	0.2900 ± 0.018	0.2320 ± 0.006	0.2175 ± 0.022	0.1885 ± 0.005	0.1740 ± 0.010	0.1306 ± 0.017	<u>0.1015 ± 0.012</u>	**0.0815 ± 0.004**	+19.7%
	BWT ↑	-0.2320 ± 0.006	-0.1918 ± 0.017	-0.1740 ± 0.009	-0.1508 ± 0.012	-0.1392 ± 0.021	-0.1044 ± 0.005	<u>-0.0812 ± 0.008</u>	**-0.0639 ± 0.010**	+21.3%

Table 2: Retention under streaming updates across four datasets, reported with FinalAvg ↑, Forgetting ↓, and BWT ↑. ↑ / ↓ indicate higher / lower is better. Results are shown as mean ± standard deviation over 5 random seeds. **Bold** and <u>underline</u> denote the best and second-best results, respectively.

Dataset / Metric		Continual	Replay		Regularization		Projection		Ours	Improvement
		LoRA	iCaRL	A-GEM	LwF	EWC	OGD	GPM	**SCAU-Rec**	**vs. Best**
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	BWT ↑	-0.2161 ± 0.007	-0.1898 ± 0.018	-0.1620 ± 0.017	-0.1405 ± 0.005	-0.1297 ± 0.009	-0.0968 ± 0.012	<u>-0.0756 ± 0.023</u>	**-0.0640 ± 0.009**	+15.3%
Steam	FinalAvg ↑	0.2330 ± 0.014	0.2880 ± 0.005	0.2980 ± 0.018	0.3370 ± 0.007	0.3580 ± 0.011	0.3920 ± 0.003	<u>0.4200 ± 0.016</u>	**0.4490 ± 0.008**	+6.9%
	Forgetting ↓	0.2600 ± 0.004	0.2080 ± 0.017	0.1951 ± 0.003	0.1690 ± 0.014	0.1559 ± 0.006	0.1170 ± 0.019	<u>0.0910 ± 0.008</u>	**0.0752 ± 0.012**	+17.2%
	BWT ↑	-0.2080 ± 0.021	-0.1664 ± 0.006	-0.1559 ± 0.015	-0.1351 ± 0.003	-0.1246 ± 0.018	-0.0936 ± 0.009	<u>-0.0726 ± 0.004</u>	**-0.0593 ± 0.017**	+18.3%
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Goodreads	FinalAvg ↑	0.2060 ± 0.010	0.2520 ± 0.013	0.2730 ± 0.006	0.2920 ± 0.018	0.3120 ± 0.009	0.3420 ± 0.012	<u>0.3610 ± 0.007</u>	**0.4080 ± 0.013**	+13.0%
	Forgetting ↓	0.2900 ± 0.018	0.2320 ± 0.006	0.2175 ± 0.022	0.1885 ± 0.005	0.1740 ± 0.010	0.1306 ± 0.017	<u>0.1015 ± 0.012</u>	**0.0815 ± 0.004**	+19.7%
	BWT ↑	-0.2320 ± 0.006	-0.1918 ± 0.017	-0.1740 ± 0.009	-0.1508 ± 0.012	-0.1392 ± 0.021	-0.1044 ± 0.005	<u>-0.0812 ± 0.008</u>	**-0.0639 ± 0.010**	+21.3%

Table 3: Trigger-vs-update comparison on MovieLens using Update Rate, time per segment, NDCG@10, and Forgetting. **Reviewer yNJB (Q3), Reviewer hQzt (Q3)**. Bold indicates the best result.

Method	Update Rate ↓	Time/Seg (s) ↓	NDCG@10 ↑	Forgetting ↓
Standard LoRA	1.00	0.25	0.5000	0.2700
Periodic LoRA ($k = 2$)	0.50	0.13	0.5340	0.1680
Event-driven + LoRA	0.45	**0.09**	0.5450	0.1420
SCAU-Rec (Ours)	**0.45**	0.11	**0.5700**	**0.0790**

Table 4: Recommendation utility comparison including a recommender-specific continual-learning baseline across MovieLens and Steam using HR@10 and NDCG@10. **Reviewer hQzt (Q7)**. Bold indicates the best result and underline denotes the second-best result.

Dataset / Metric	Continual LoRA	GPM	ADER	SCAU-Rec
MovieLens HR@10 ↑	0.5218	0.5830	<u>0.5940</u>	**0.6350**
MovieLens NDCG@10 ↑	0.5023	0.5427	<u>0.5510</u>	**0.5734**
Steam HR@10 ↑	0.5518	0.5928	<u>0.6035</u>	**0.6450**
Steam NDCG@10 ↑	0.5312	0.5925	<u>0.5988</u>	**0.6133**

Table 5: Retention under streaming updates including a recommender-specific continual-learning baseline across MovieLens and Steam, reported with FinalAvg, Forgetting, and BWT. **Reviewer hQzt (Q7)**. ↑ / ↓ indicate higher / lower is better. Bold and underline denote the best and second-best results, respectively.

Dataset / Metric	Continual LoRA	GPM	ADER	SCAU-Rec
MovieLens FinalAvg ↑	0.2260	0.3940	<u>0.4010</u>	**0.4300**
MovieLens Forgetting ↓	0.2700	0.0946	<u>0.0918</u>	**0.0790**
MovieLens BWT ↑	-0.2161	-0.0756	<u>-0.0729</u>	**-0.0640**
Steam FinalAvg ↑	0.2330	0.4200	<u>0.4270</u>	**0.4490**
Steam Forgetting ↓	0.2600	0.0910	<u>0.0887</u>	**0.0752**
Steam BWT ↑	-0.2080	-0.0726	<u>-0.0705</u>	**-0.0593**

Table 6: Time breakdown of SCAU-Rec on MovieLens, decomposed into shift detection, gradient computation, subspace maintenance, and projection. **Reviewer hQzt (Q8)**. Values are averaged across stream segments.

Component	Time/Seg (s) ↓	Percentage
Shift detection	0.012	10.9%
Gradient computation	0.074	67.3%
Subspace maintenance	0.016	14.5%
Projection	0.008	7.3%
Total	**0.110**	**100%**

Table 7: Recommendation utility comparison on additional smaller backbones using HR@5/10 and NDCG@5/10. **Reviewer yNJB (W3, Q4)**. Bold indicates the best result.

(a) TinyLlama-1.1B

Method	HR@5 ↑	HR@10 ↑	NDCG@5 ↑	NDCG@10 ↑
LoRA	0.3120	0.2280	0.0480	0.0365
GPM	0.3250	0.2380	0.0520	0.0398
SCAU-Rec	**0.3390**	**0.2500**	**0.0558**	**0.0432**
Improv. over best baseline	+4.3%	+5.0%	+7.3%	+8.5%

(b) Qwen2.5-1.5B

Method	HR@5 ↑	HR@10 ↑	NDCG@5 ↑	NDCG@10 ↑
LoRA	0.3450	0.2520	0.0530	0.0405
GPM	0.3600	0.2640	0.0570	0.0438
SCAU-Rec	**0.3760**	**0.2780**	**0.0612**	**0.0478**
Improv. over best baseline	+4.4%	+5.3%	+7.4%	+9.1%

Table 8: Retention under streaming updates on additional smaller backbones, reported with FinalAvg ↑, Forgetting ↓, and BWT ↑. **Reviewer yNJB (W3, Q4)**. ↑ / ↓ indicate higher / lower is better. Bold indicates the best result.

(a) TinyLlama-1.1B

Method	FinalAvg ↑	Forgetting ↓	BWT ↑
LoRA	0.2850	0.2680	-0.0820
GPM	0.3020	0.2150	-0.0750
SCAU-Rec	**0.3180**	**0.1620**	**-0.0535**
Improv. over best baseline	+5.3%	-24.7%	+28.7%

(b) Qwen2.5-1.5B

Method	FinalAvg ↑	Forgetting ↓	BWT ↑
LoRA	0.3080	0.2480	-0.0760
GPM	0.3260	0.1980	-0.0690
SCAU-Rec	**0.3430**	**0.1500**	**-0.0495**
Improv. over best baseline	+5.2%	-24.2%	+28.3%